COURSE TITLE : SURVEYING FOR ARCHITECTURE
COURSE CODE : 3182
COURSE CATEGORY : B
PERIODS/WEEK : 6
PERIODS/SEMESTER : 90
CREDITS : 4

TIME SCHEDULE

Module	Topic	Periods
1	Surveying instruments	18
2	Leveling	30
3	Theodolite Surveying	24
4	Total Station	18
TOTAL		**90**

Sl.	Sub	Student will be able to
1	1	Understand the purpose of surveying
	2	Apply the principles of chain survey to calculate areas
	3	Understand the principles of plane table survey
2	**1**	Apply the principle of Compass survey to determine inaccessible distance
	2	Apply the principle of leveling
	3	Classify leveling
3	**1**	Apply the principle of contouring
	2	Apply the principles of theodolite traversing
	3	Apply the principle of trigonometry for determining the elevation of stations get exposure to modern surveying equipments
4	**1**	To apply the principles of Total Station

MODULE I

1.1.0 Understand the purpose of surveying

1.1.1 State the type of survey used for a given engineering purpose.

1.2.0 Apply the principles of chain survey to calculate areas

1.2.1 List the functions of chain, tape and cross-staff.

1.2.2 List the operations involved in chain survey to make site plan.

1.2.3 Make a survey plan from given field notes.

1.3.0 Understand the principles of plane table survey

. 1.3.1 Identify the functions of accessories of plane table.

1.3.2 List the operations to set up and orient the plane table.

1.3.2 List the advantages and disadvantages.

1.4.0 Apply the principle of Compass survey to determine inaccessible distance

1.4.1 Identify the parts and their functions of a prismatic compass.

1.4.2 List the operations involved in the field to take various bearings.

1.4.2 Determine the inaccessible distance by using prismatic compass.

MODULE II

2.1.0 Apply the principle of leveling

2.1.1 Identify the parts and functions of dumpy level and telescopic leveling staff.

2.1.2 List the steps involved in performing the temporary adjustments of leveling Instrument.

2.1.3 Given a list of observations, record them in field book form.

2.1.4 Compute the reduced levels of stations from field book.

2.2.0 Classify leveling

2.2.1 List the steps involved in classification of leveling.

2.3.0 Apply the principle of contouring.

2.3.1 Prepare contour plans from given field notes.

MODULE III

3.1.0 To apply the principles of theodolite traversing

3.1.1 To identify the theodolite, their parts and function.

3.1.2 To know the temporary adjustments and terms.

3.1.3 To list the fundamental lines and their relationship.
3.1.4 To study horizontal angle measurement by repetition and reiteration method.
3.1.5 To list the steps involved in setting out angles using a theodolite.
3.1.6 To explain the method of conducting traverse survey in the field using the theodolite.
3.1.7 To compute the co-ordinates.

3.2.0 To apply the principle of trigonometry for determining the elevation of stations

3.2.1 To study vertical angle measurements.
3.2.2 To explain the principle of trigonometrically leveling.
3.2.3 To compute the elevations using trigonometrically leveling.

MODULE IV

4.1.0 To get exposure to modern surveying equipments

4.1.1 To identify the equipments like electronic theodolite, distomat, laser level and GPS
4.1.2 To understand remote sensing in architecture.
4.1.3 Understand the application of GIS in architecture.

4**.2.0 To apply the principles of Total Station**

4.2.1 Identify the total station, their features and technical terms.
4.2.2 Study the method of operation.
4.2.3 Make a contour map of an area.
4.2.4 Determine height of building.
4.2.5 Determine horizontal distance between points.
4.2.6 Describe data down loading.
4.2.7 Describe the procedure to prepare drawing by using total station.

COURSE CONTENTS

MODULE I

Purpose of surveying - types of survey - chain survey - calculate areas - functions of chain, tape and cross-staff - operations involved in chain survey to make site plan - survey plan from given field notes - principles of plane table survey - functions of accessories of plane table - operations to set up and orient the plane table - advantages and disadvantages - principle of Compass survey - determine inaccessible distance - the parts and their functions of a prismatic compass-operations involved in the field to take various bearings - inaccessible distance by using prismatic compass.

MODULE II

Principle of leveling **-** parts and functions of dumpy level and telescopic leveling staff - steps involved in performing the temporary adjustments of leveling Instrument - record observations in field book - compute the reduced levels of stations from field book – Classifications of leveling - steps involved in classification of leveling - principle of contouring - contour plans from given field notes.

MODULE III

Principles of theodolite traversing - their parts and function -temporary adjustments and terms - fundamental lines and their relationship - horizontal angle measurement by repetition and reiteration method - steps involved in setting out angles using a theodolite - method of conducting traverse survey in the field using the theodolite - compute the co-ordinates - principle of trigonometry for determining the elevation of stations -vertical angle measurements -principle of trigonometric leveling - compute the elevations using trigonometric leveling.

MODULE IV

Modern surveying equipments - electronic theodolite – distomat - laser level – GPS - remote sensing in architecture -application of GIS in architecture - principles of Total Station -features and technical terms -method of operation - contour map of an area - Determine height of building - horizontal distance between points - data down loading - prepare drawing by using total station.

REFERENCE BOOKS

1. T.P.Kanetkar & Kulkarni : Surveying and Levelling (VolI&Vol II) ; Jain book depot
2. B.C.Punmia : Surveying – II ; Laxmipublications
3. K.R Arora : Surveying - ; Standard Book House
4. P.B.Shahai : A textbook of surveying [Vol.I and Vol. II ; Oxford and IBH Publishing Co.
5. Patel .A.N : Remote Sensing Principles & Applications ; Scientific Publishers
6. Lillesand : Remote sensing and image interpretation ; Wiley
7. SathishGopi : Advanced Surveying ; Pearson
8. CL Kocher : Text book of Surveying ; Dhanpat Rai Publishing Co (p)
9. N N Basak : Surveying ; TataMcgraw Hill
10. S.S.Bhavikatti : Surveying and leveling (Vol. I &II) ; I.K.International publishing house
11. NITTTR, Chennai : AICTE Continuing Education module on Geographical information systems